x86 Processor Memory Management

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Course ID: CSE - 4503
Course Title: Microprocessors and Assembly Language
Department of Computer Science and Engineering (CSE),
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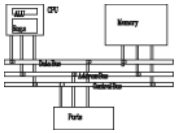
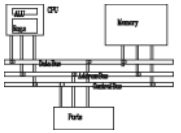
Lecturer Reference

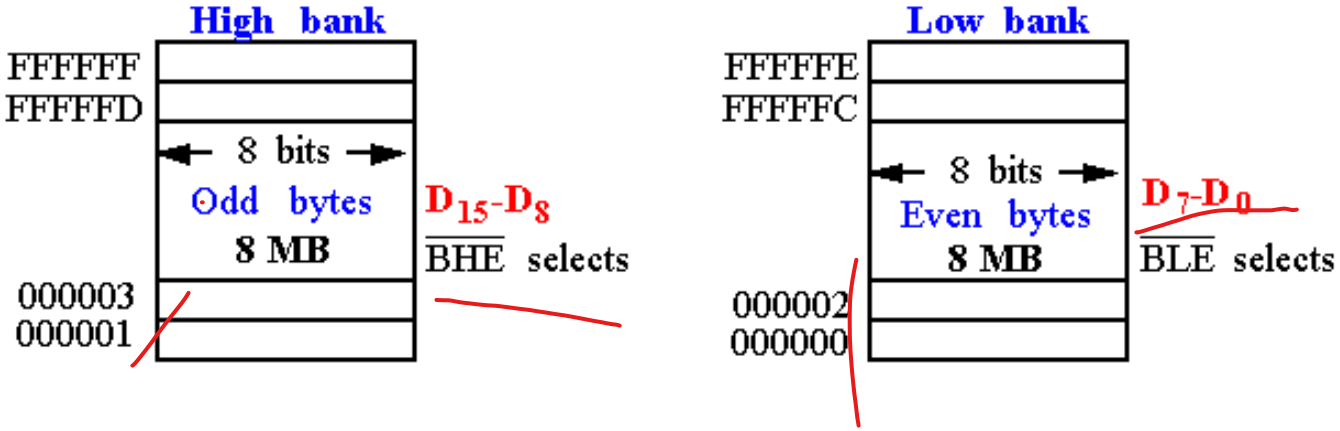


- Lecture Material:
  - Intel x86 Architecture: *Computer Organization and Assembly Languages, Yung-Yu Chuang*
  - *with slides by Kip Irvine*

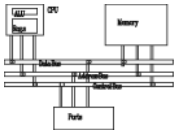
Memory Banks of 8086



- **ODD Bank and EVEN BANK**
- If a 16-bit data is stored in a memory location started with even address and ends at odd address, then the 8086 processor can read the data in one cycle.
- If a 16-bit data is stored in a memory location started with a odd address and ends at even address, then the 8086 processor can read the data in two cycles.
- If 8-bit data is stored in either even or odd memory location then the 8086 processor can read it in one cycle.



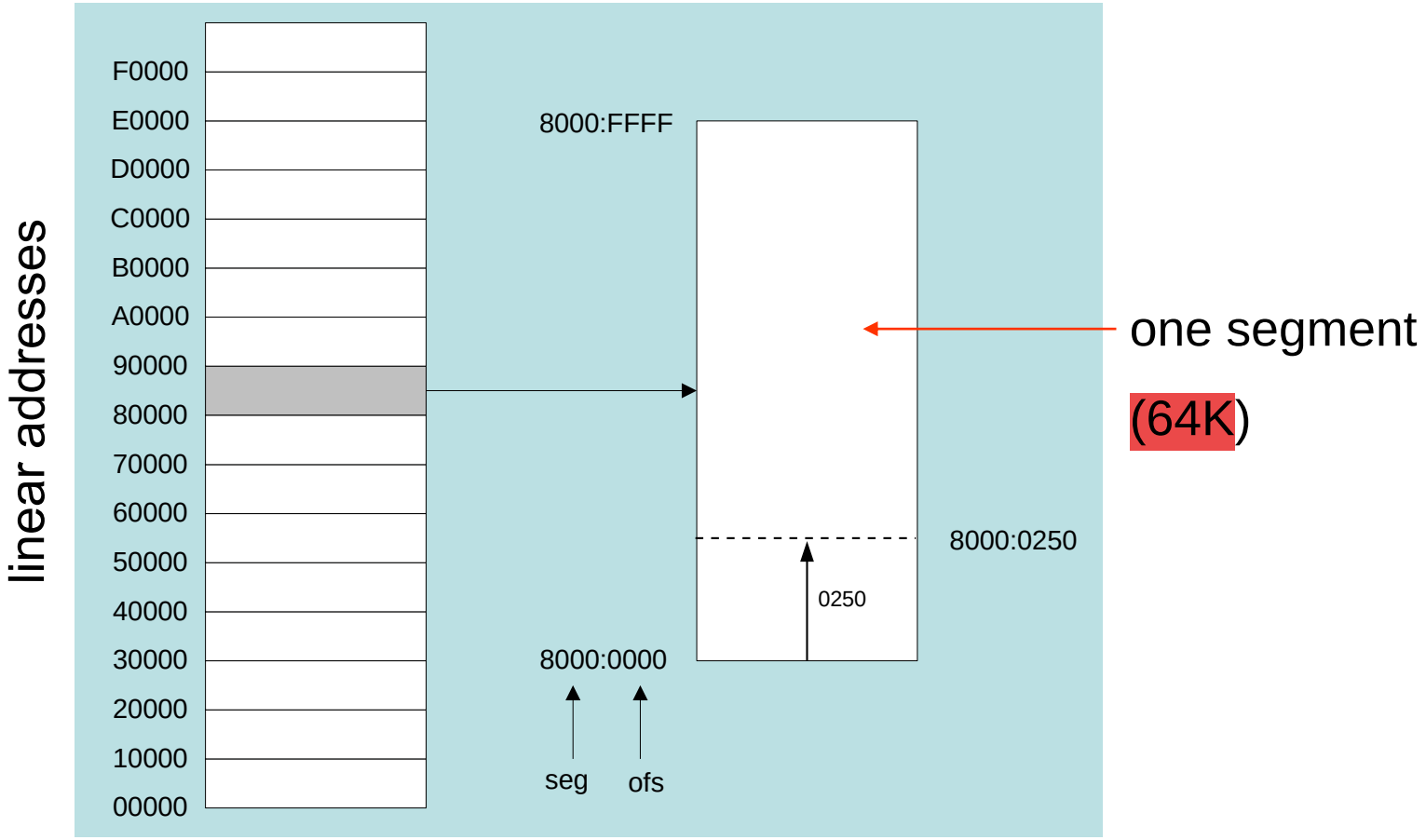
Real-address mode (in 8086)



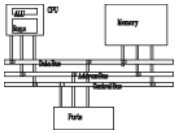
- 1 MB RAM maximum addressable (20-bit address)
- Application programs can access any area of memory
- Single tasking
- Supported by MS-DOS operating system

Real-address Mode: Segmented memory

Segmented memory addressing: absolute (linear) address is a combination of a 16-bit segment value added to a 16-bit offset



Real-address Mode: Calculating linear addresses

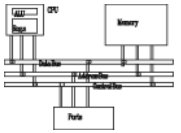


- Given a segment address, multiply it by 16 (add a hexadecimal zero), and add it to the offset
- Example: convert 08F1:0100 to a linear address

Adjusted Segment value:	**0**	**8**	**F**	**1**	**0**
Add the offset:		**0**	**1**	**0**	**0**
Linear address:	**0**	**9**	**0**	**1**	**0**

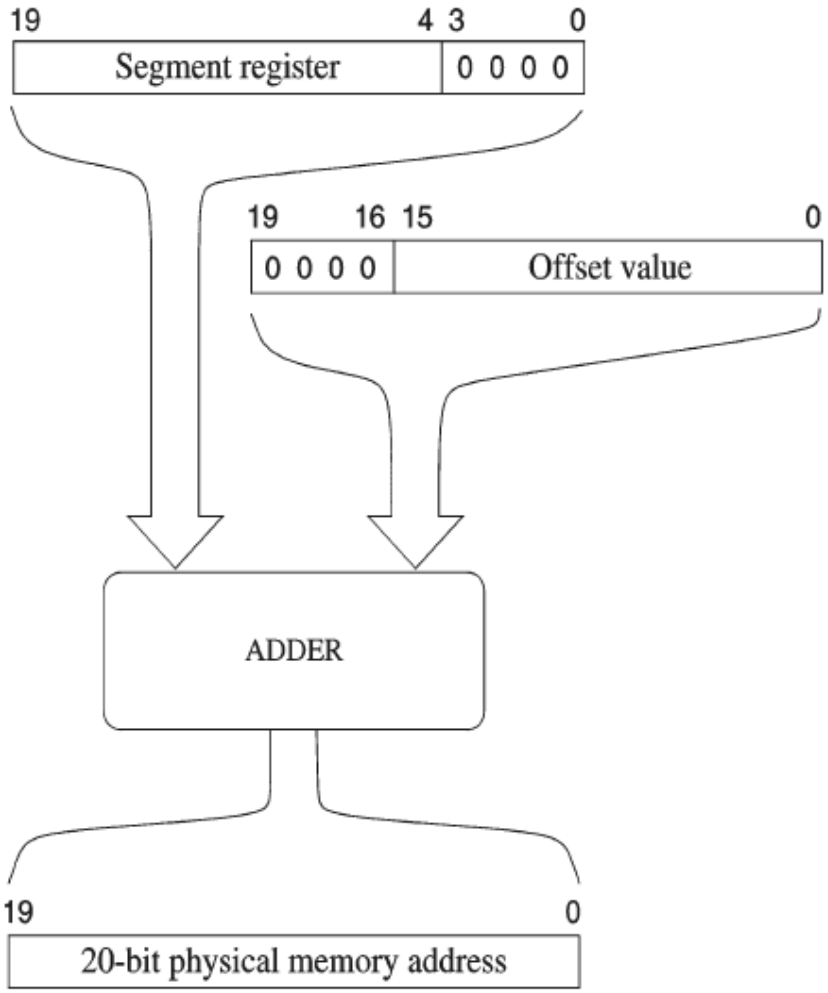
- **A typical program has three segments**: code, data and stack. Segment registers CS, DS and SS are used to store them separately.

Real-address Mode: Calculating linear addresses

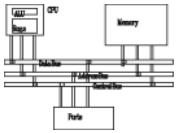


- Conversion from logical to physical addresses

```
  11000  (add 0 to base)
+   450  (offset)
-------
  11450  (physical address)
```



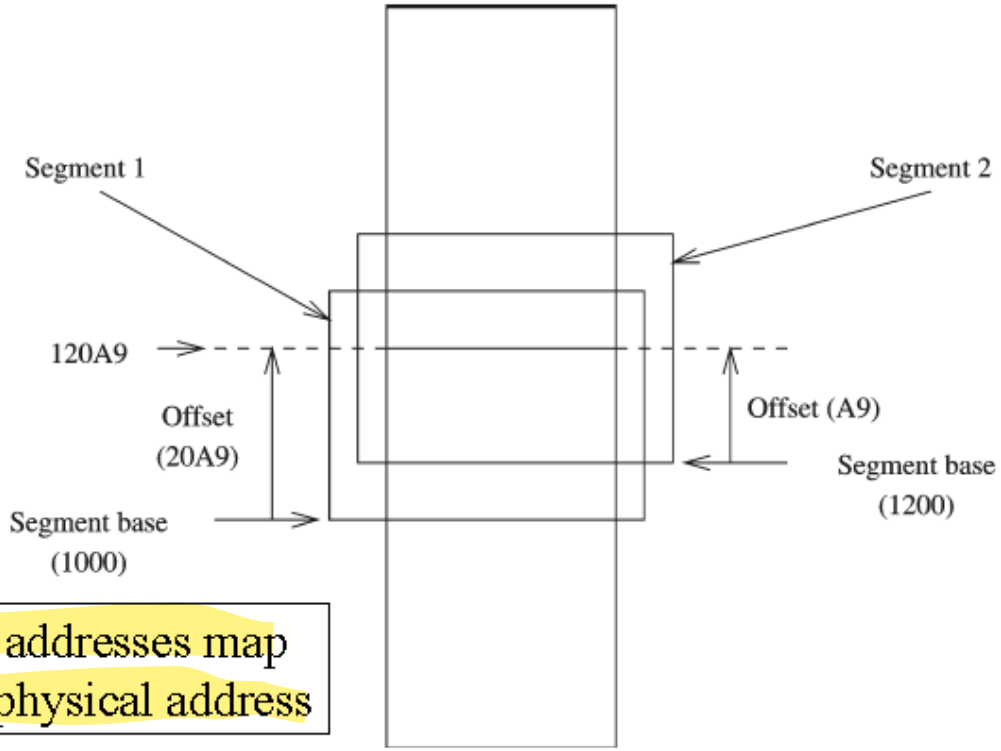
Real-address Mode: Example



What linear address corresponds to the segment/offset address 028F:0030?

028F0 + 0030 = 02920

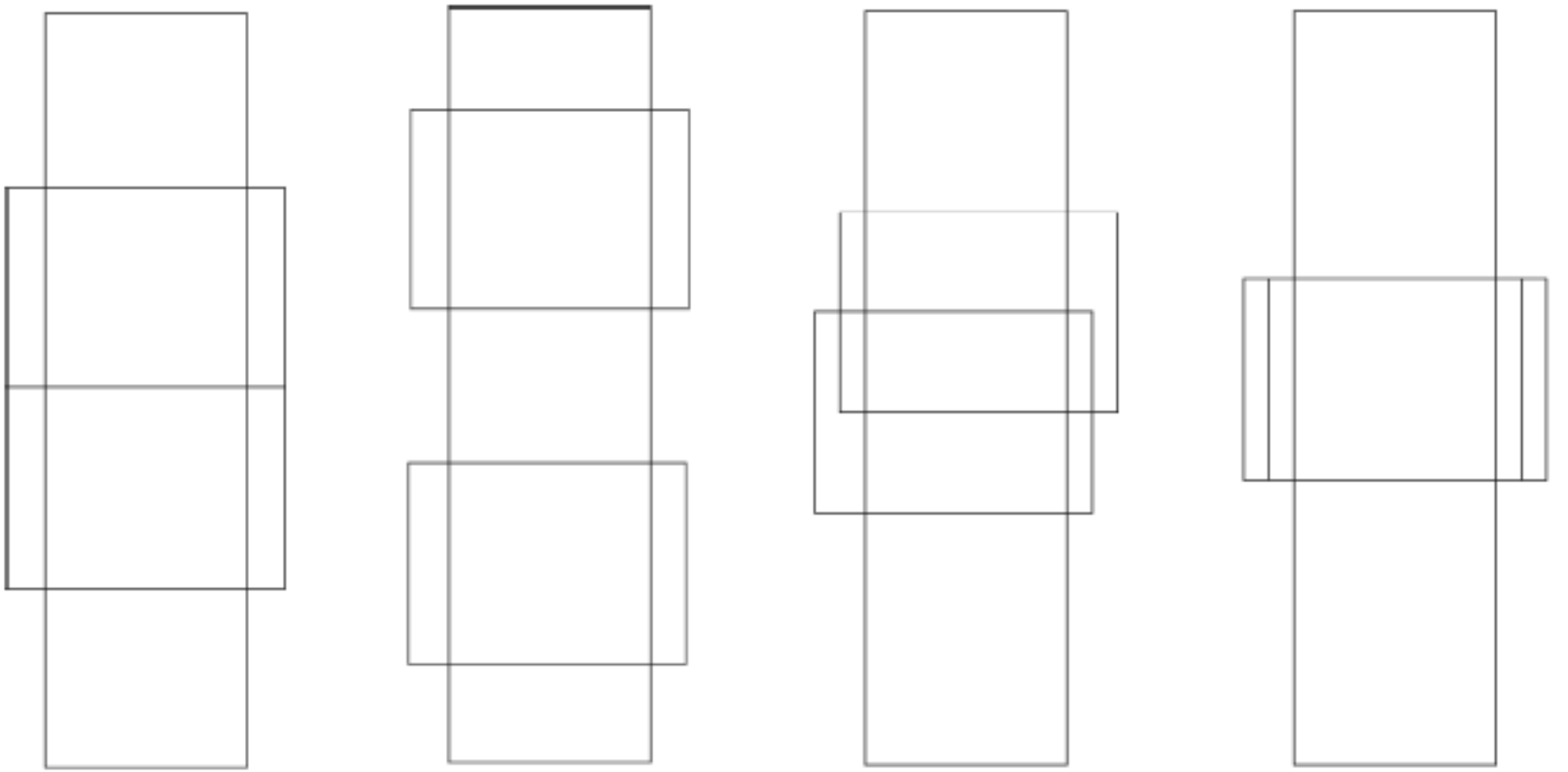
Always use hexadecimal notation for addresses.



Two logical addresses map to the same physical address

Real-address Mode: Example

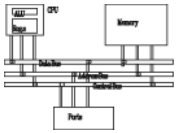
- Segment Overlapping



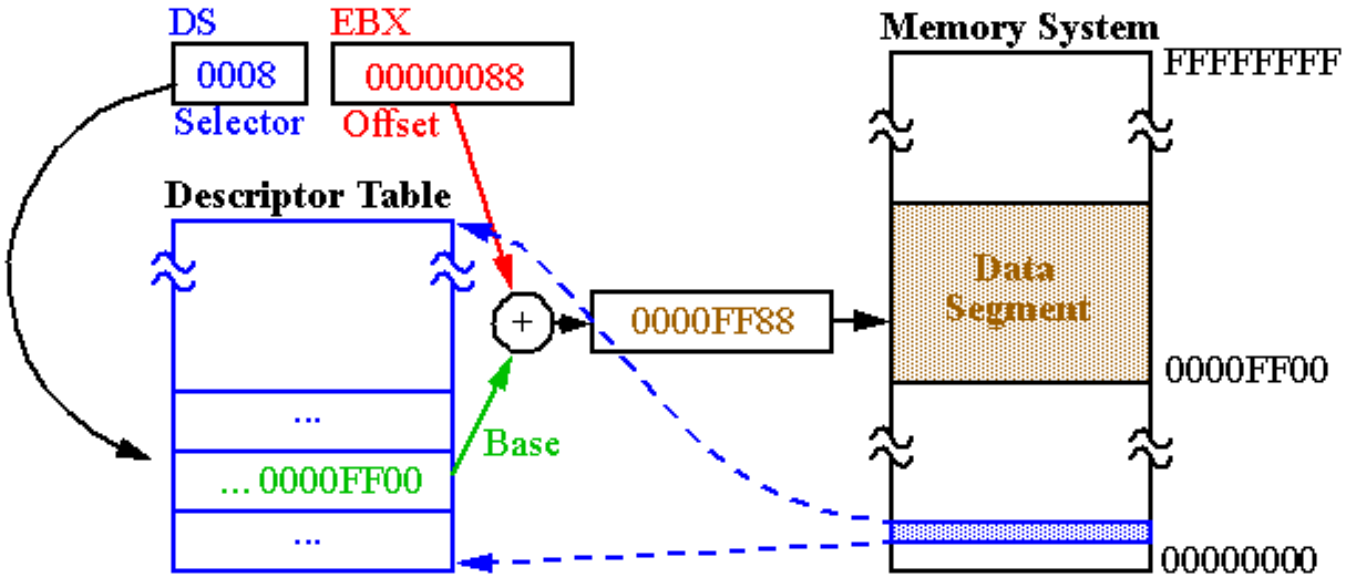
(a) Adjacent (b) Disjoint (c) Partially overlapped (d) Fully overlapped

EMU 8086 uses Fully Overlapped Segment

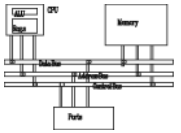
Protected mode (in 80286 and Upper)



- 80286 processor with 16 MB addressable RAM (24-bit address bus) (000000 to FFFFFFh)
- 80386 and upper processors with 4 GB addressable RAM (32-bit address bus) (00000000 to FFFFFFFFh)
- **Each program** assigned with a memory partition which is protected from other programs
- Designed for multitasking
- Supported by Linux & MS-Windows

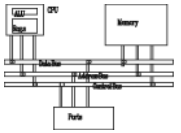


Protected mode



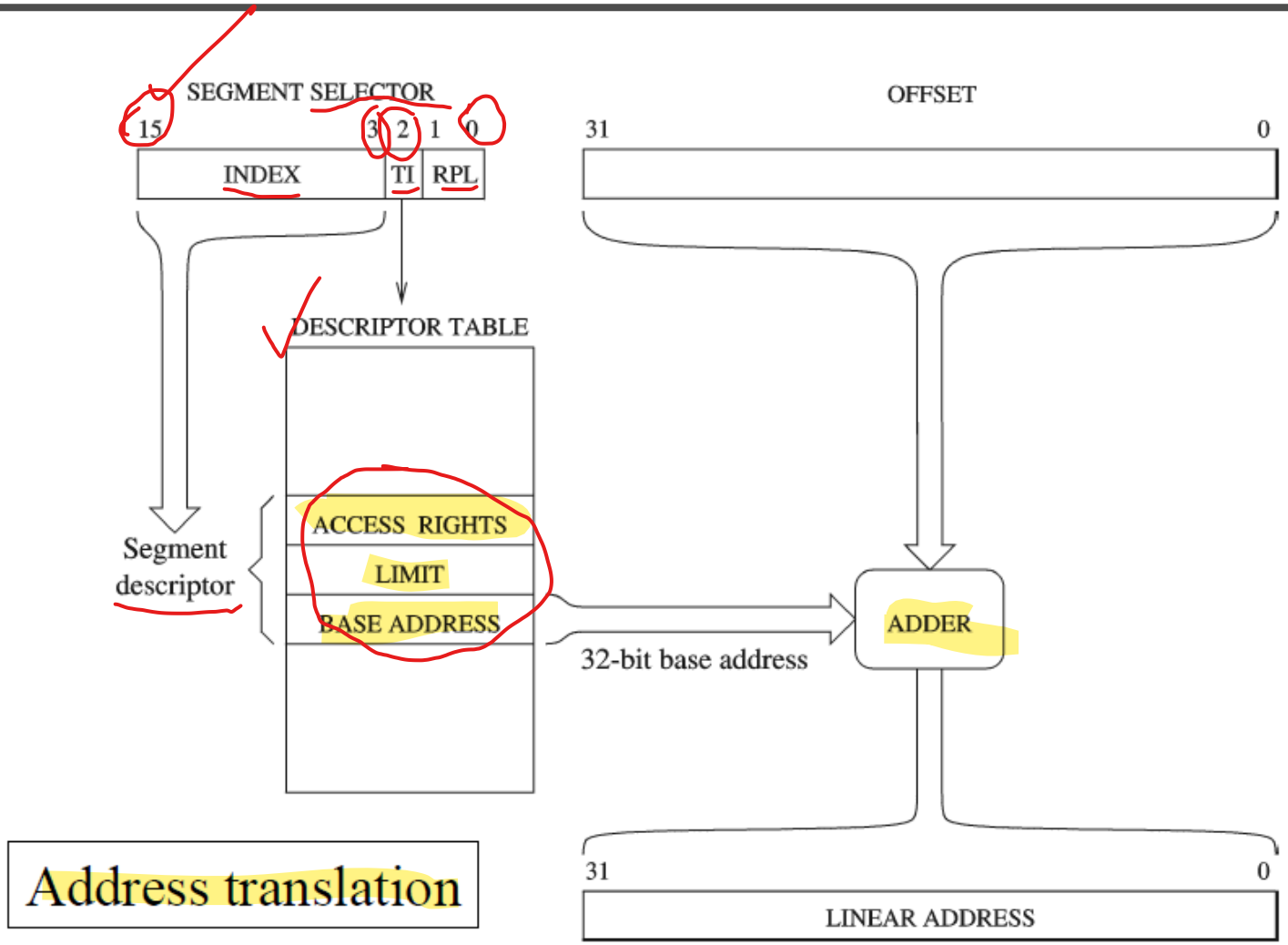
- Started with 80286 up to Pentium, all processors use 2 modes for memory address management
  - Real mode
    - » Uses 16-bit addresses
    - » Runs 8086 programs
    - » Pentium acts as a faster 8086
  - Protected mode
    - » 32-bit mode
    - » Native mode of Pentium
    - » Supports segmentation and paging

Protected mode (Types of Address Space)



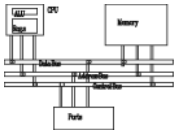
- **Logical/Virtual Address:**
  - An address used by application program consists of 16-bit selector and 32-bit offset value.
  - In a flat memory model, 16-bit selector is loaded in CS, DS, SS and ES registers.
- **Linear Addresses:**
  - It is calculated from virtual address by **segmentation unit**.
  - The base of the segment refer to by the selector is added with the offset, generating a 32-bit linear address.
  - **When paging is disabled, Linear Address = Physical Address**
- **Physical Address:**
  - The address value appears on the address bus of a microprocessor during a memory read/write operation
  - Physical address calculated from linear addresses through **paging unit.**
  - Linear address is used as an index into the page table, where the CPU locates the corresponding physical address.

Protected Mode: Calculating Linear Addresses



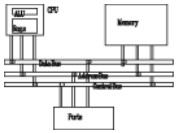
Address translation

Protected Mode: Calculating Addresses



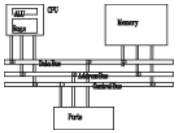
- Index
  - Selects a descriptor from one of two descriptor tables
    - Local
    - Global
- Table Indicator (TI)
  - Select the descriptor table to be used
    - 0 = Local descriptor table
    - 1 = Global descriptor table
- Requestor Privilege Level (RPL)
  - Privilege level to provide protected access to data
    - Smaller the RPL, higher the privilege level

Protected mode



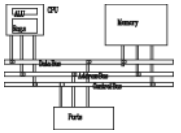
- Supports sophisticated segmentation
- Segment unit translates 32-bit logical address to 32-bit linear address
- Paging unit translates 32-bit linear address to 32-bit physical address
  - If no paging is used
    - Linear address = physical address

Protected mode



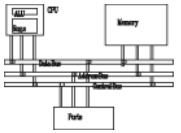
- In this mode there is a Segment Descriptor Table
- Typical Program structure follows:
  - Code, Data, and Stack areas
  - CS, DS, SS segment descriptors
  - Global Descriptor Table (GDT)
  - Local Descriptor Table (LDT)

Protected mode (Address Translation)



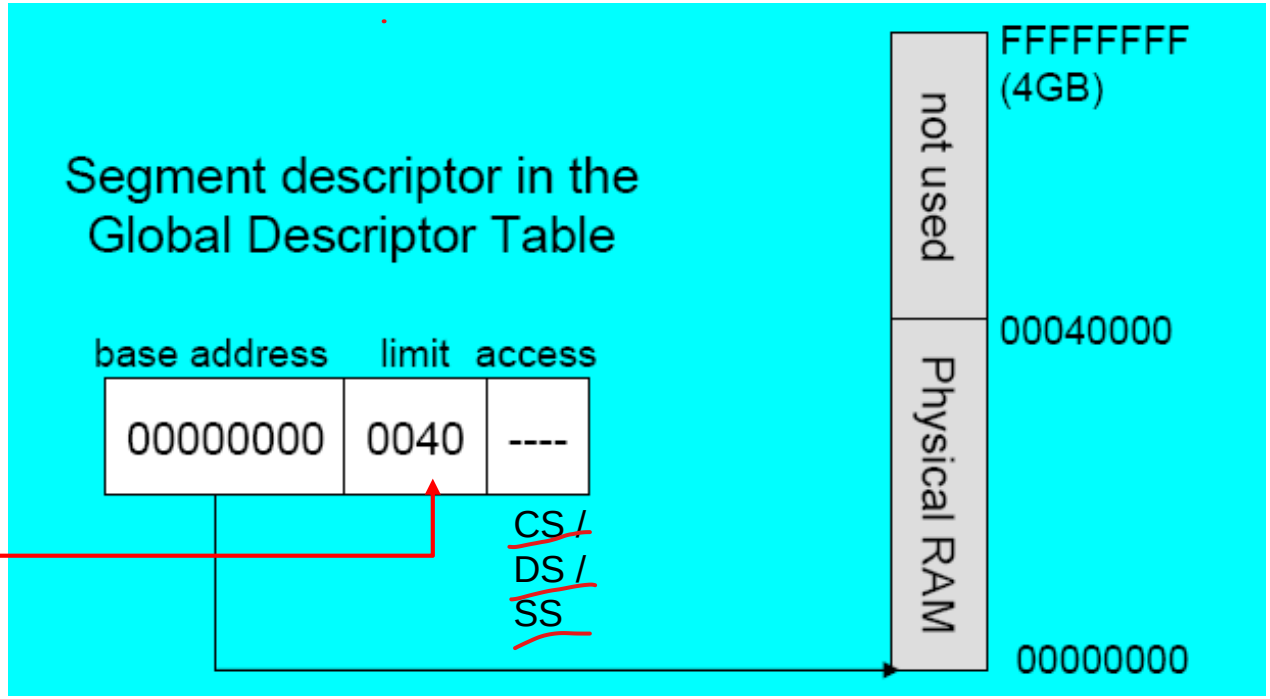
- 80386 and later microprocessors transforms logical address into physical address in 2 steps:
  - Segment translation
  - Page translation
- These translations are done in a way that it is not visible to the application programmers or users

Flat segmentation model



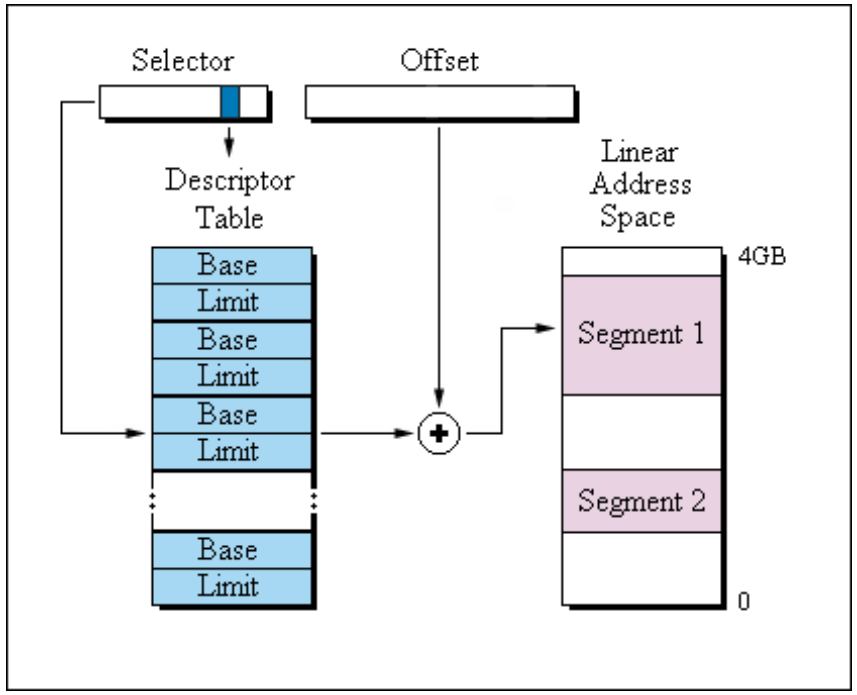
- All segments are mapped to the entire 32-bit physical address space, at least two, one for data and one for code.
- All tasks can share the GDT memory.
- GDTR Register is 48-bit. LDTR is 32-bit, of which 16-bit for **segment limit** and 16-bit for **segment selector (CS/DS/SS).** The segment selector further partitioned into 13-bits to define 8192 Descriptors, each having 8 bytes size (i.e., total 64-bits = 32-bit base address + 16-bit limit + 16-bit access). So, total memory size of GDT = 8K x 8 bytes = 64 KB.
- Global Descriptor Table (GDT)

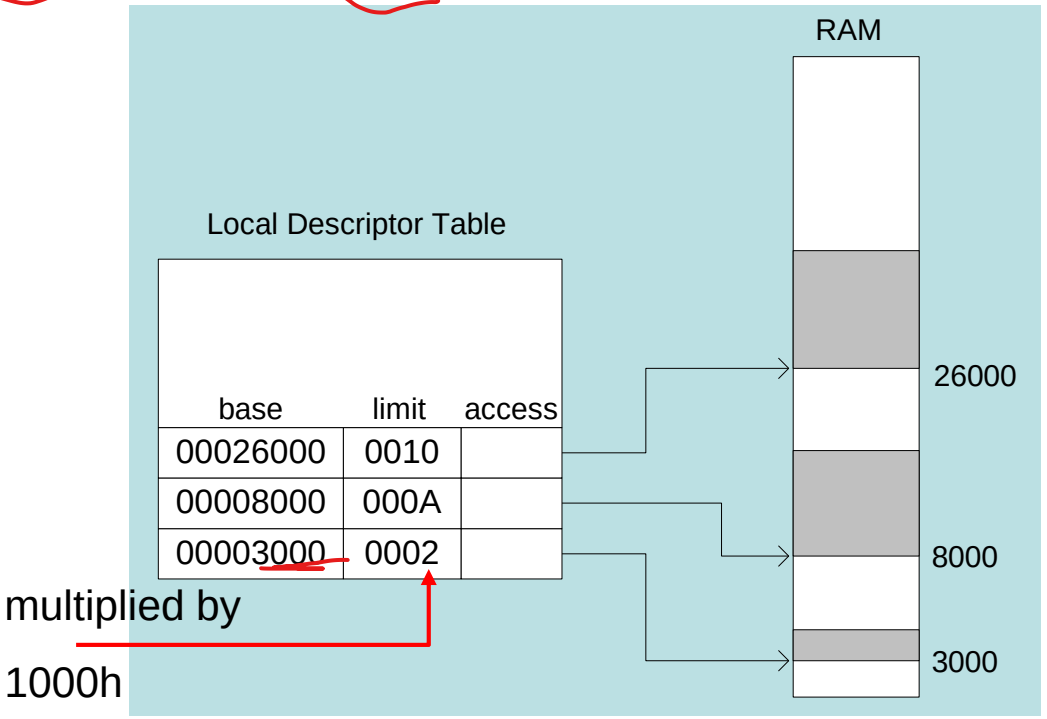
multiplied by 1000h



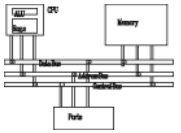
Multi-segment model

- Each program/task has a local descriptor table (LDT)
  - Holds descriptor for each segment used by the program
  - Uses any of the Descriptor of 8K GDT as LDT. So, LDT is not an independent memory table from GDT.
  - LDTR further can have max 8K Descriptors from the memory.



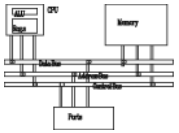


Translating Addresses

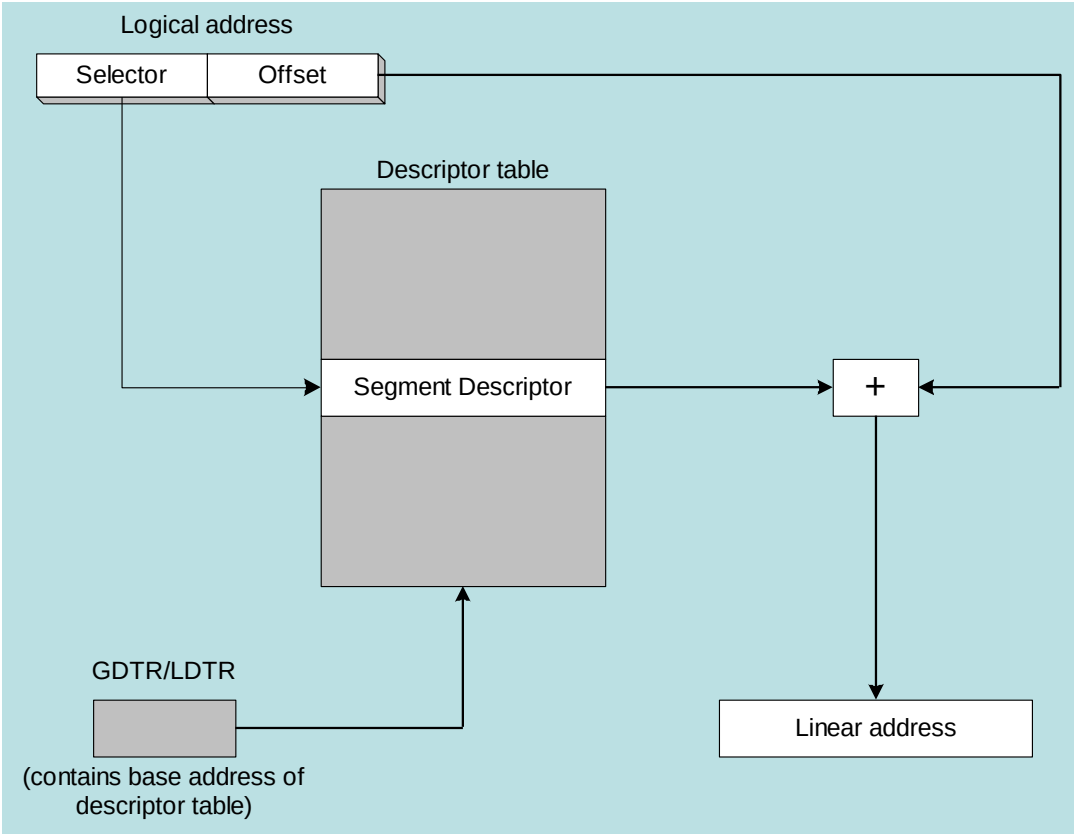


- The processor uses a one- or two-step process to convert a variable's logical address into a unique memory location.
- The first step combines a segment value with a variable's offset to create a linear address.
- The second optional step, called page translation, converts a linear address to a physical address.

Converting Logical to Linear Address

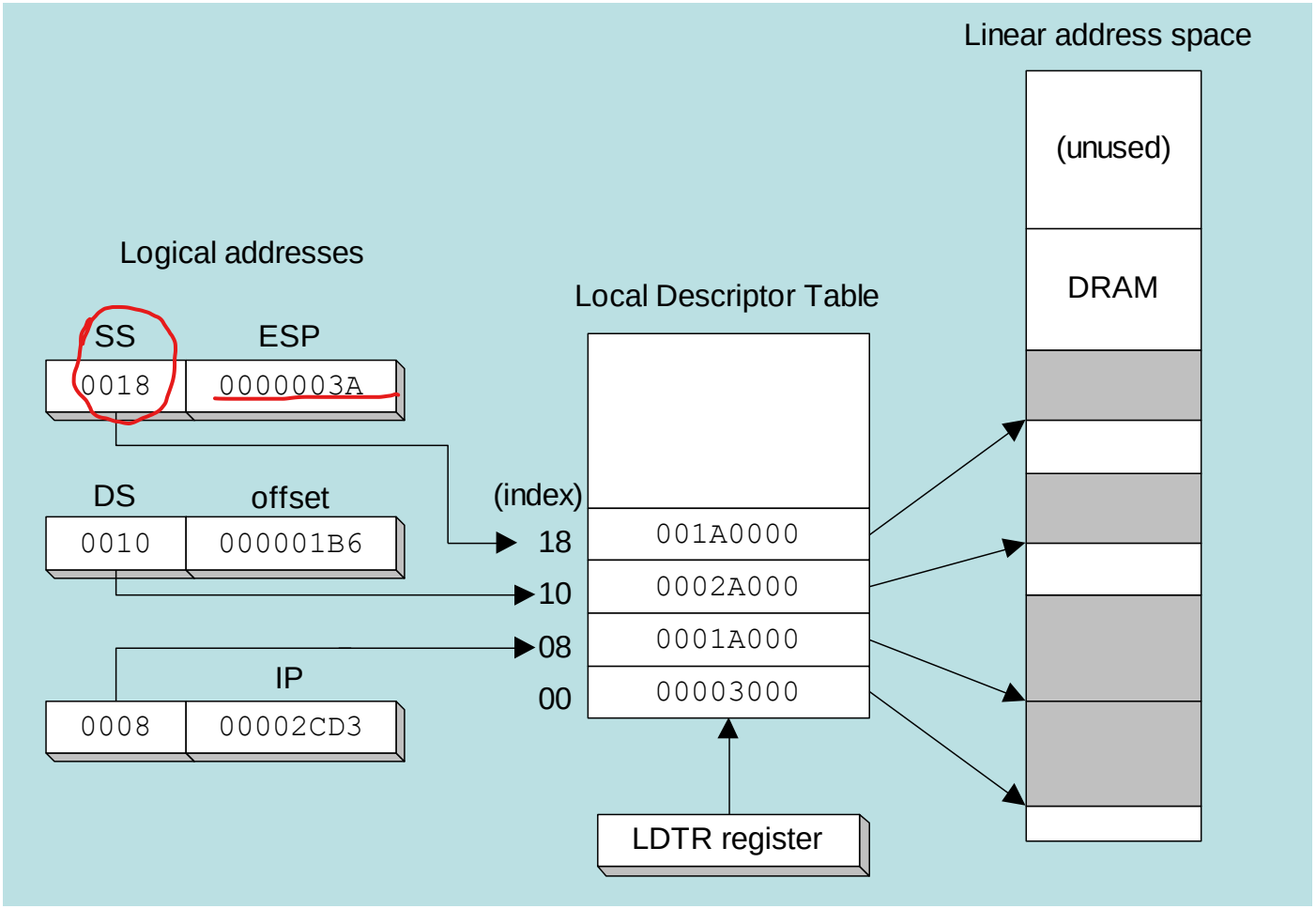


- All information about a segment is stored in a **8-byte data structure** called segment descriptor.
- The **segment selector** points to a segment descriptor, which contains the base address of a memory segment. The **32-bit offset** from the logical address is added to the segment's base address, generating a 32-bit linear address.

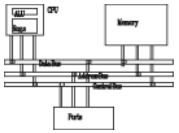


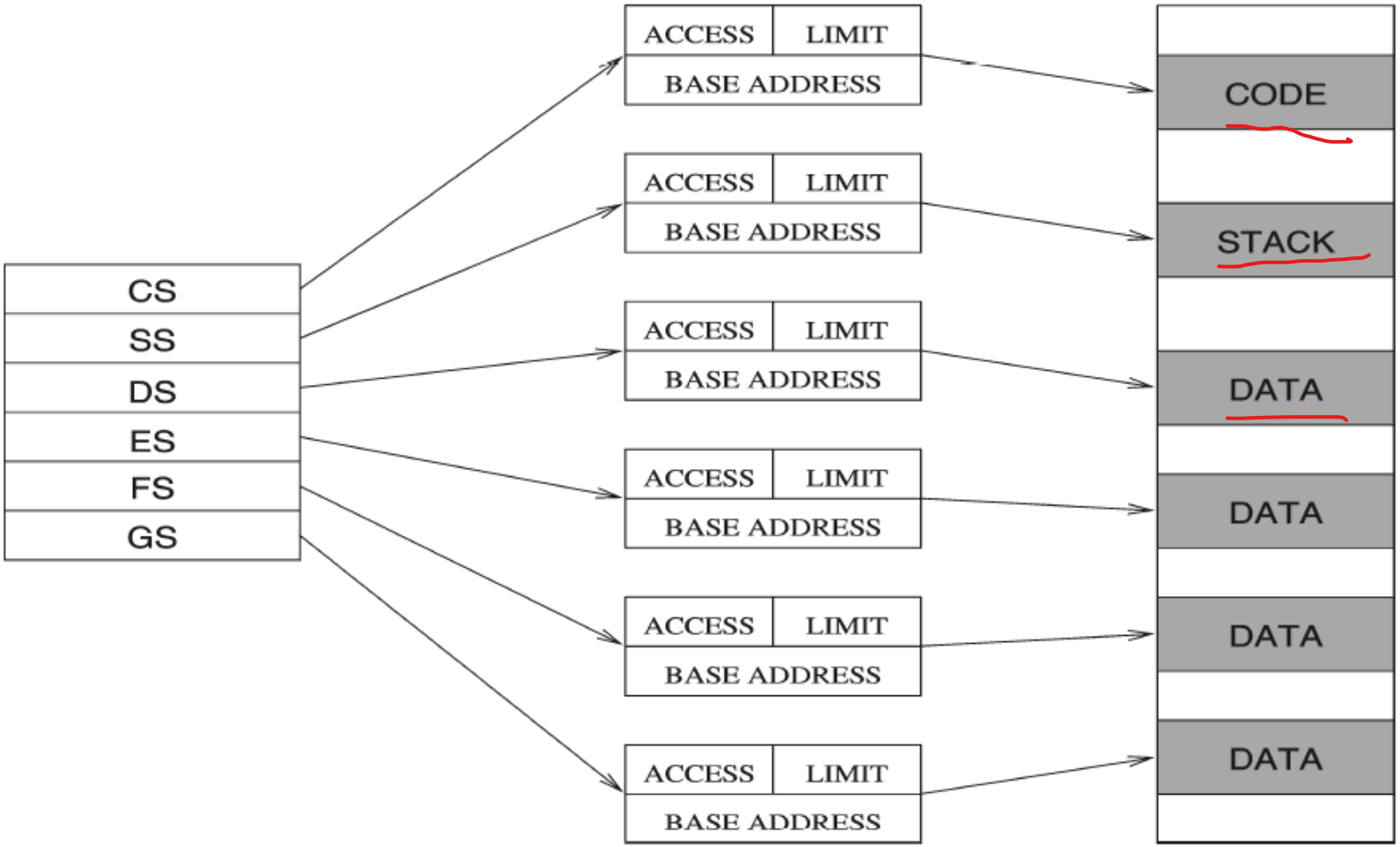
Indexing into a Descriptor Table

Each segment descriptor indexes into the program's local descriptor table (LDT). Each table entry is mapped to a linear address:

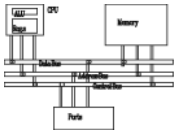


Finally …. Defining Multi-Segments



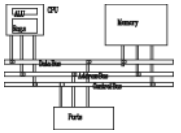


Paging



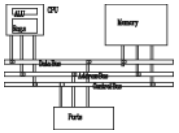
- Virtual memory uses **disk** as part of the memory, thus allowing **sum of all programs** can be larger than physical memory
- Only part of a program must be kept in **memory**, while the remaining parts are kept on **disk**.
- The **memory used by the program** is divided into small units called pages (4096-byte or 4KByte).
- As the program runs, the **processor selectively unloads inactive pages from memory** and **loads other pages** that are immediately required.

Paging

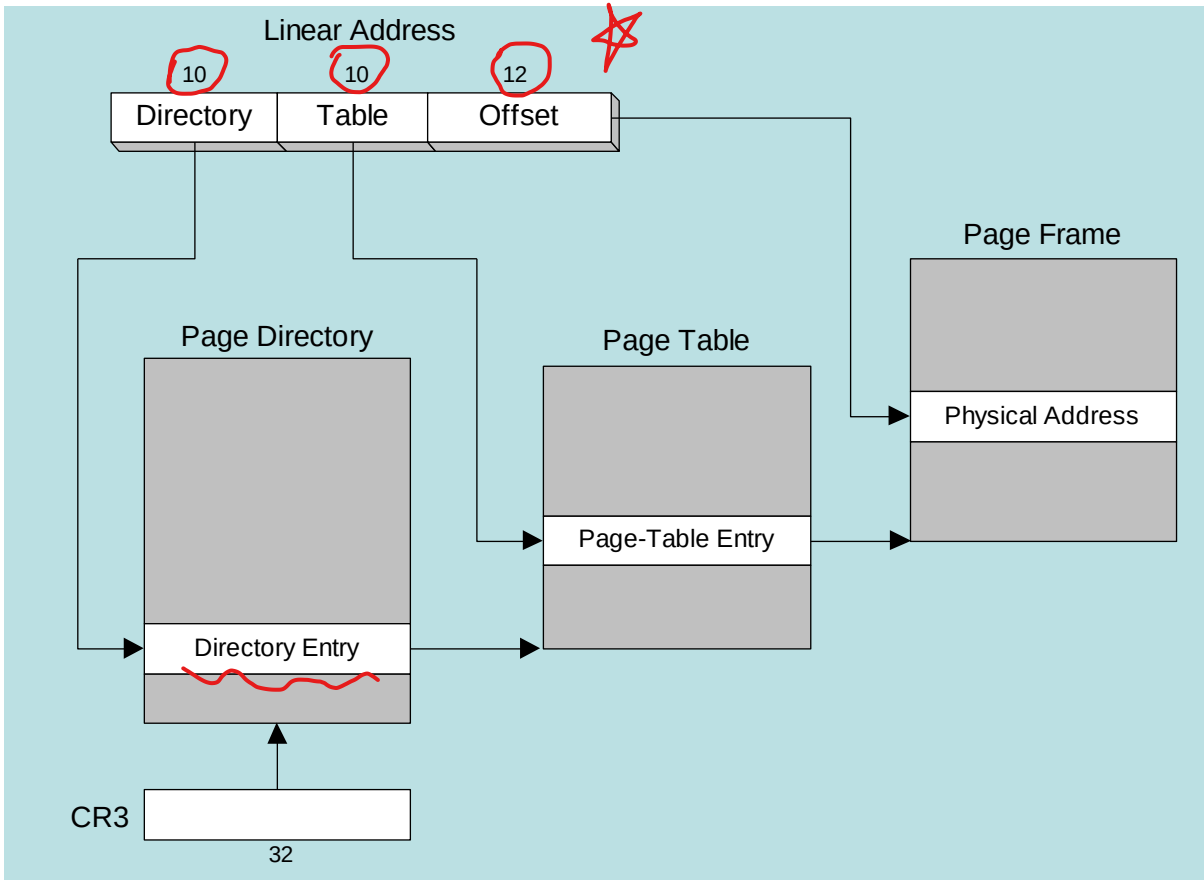


- OS maintains **page directory** and **page tables**
- **Page Translation:** CPU converts the linear address into a physical address
- **Page Fault:** Occurs when a needed page is not in memory, and the CPU interrupts the program
- **Virtual Memory Manager (VMM):** OS utility that manages the loading and unloading of pages
- OS copies the page into memory, program resumes execution

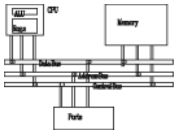
Page Translation

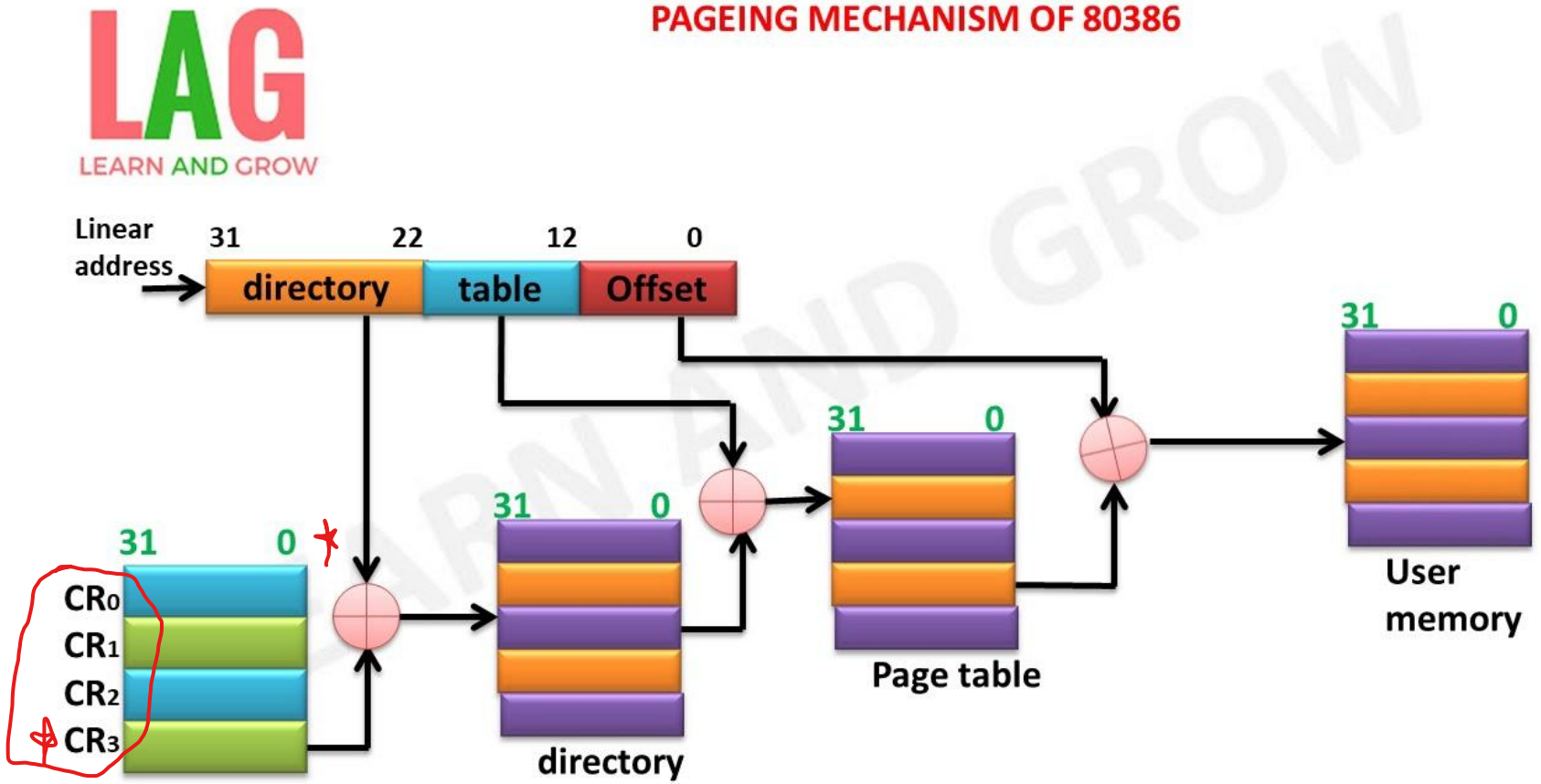


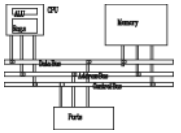
A linear address is divided into a page directory field, page table field, and page frame offset. The CPU uses all three to calculate the physical address.



Page Translation







Thank You